BOOK PROSPECTUS · OCTOBER 2026 · PREPARED FOR CENGAGE

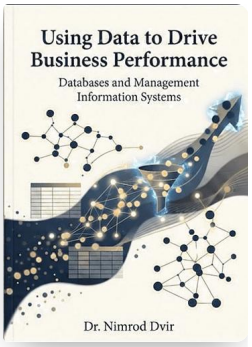
Using Data to Drive *Business Performance*

Databases and Management Information Systems

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Version 1 tested in class, summer 2026

Version 2 in use with 500 students, fall 2026



What this book is

Courseware that teaches business students to turn data into decisions: a textbook, an interactive reader, guided builds, labs, auto-graded quizzes and a semester project, built to run as one course. Students organize data, query it with SQL, check the answers, present the evidence and recommend an action, with AI as an assistant whose work they learn to check.

Data → Tables → Relationships → Queries → Analytics → Decisions

Primary course	The applied business database or MIS data-management course.
Also fits	The required business-core data or IS course, through a shorter route in the same chapters; selected modules for accounting information systems and analytics foundations.

How it is different

- **Starts with the business.** Chapter 1 opens with performance, goals and KPIs, followed by data literacy and organizing information.
- **Ends in a decision.** Analytics, visualization and a full Power BI chapter lead to a decision memo.
- **Uses a database before designing one.** Students query a supplied database, then model and build their own.
- **Makes checking a habit.** From Chapter 5, students accept, revise or reject AI-assisted work and explain why.

What exists today

Seven years of teaching this course to **more than 4,000 students** shaped the book. Version 1 was tested in class in **summer 2026**; version 2 is in use with **500 students in fall 2026**. The full textbook is live as an interactive edition, with a Let's Build, a lab and an auto-graded quiz in every chapter, an instructor dashboard, two running cases and an Access route.

Five studies behind the proposal

375,418
U.S. business bachelor's degrees, 2021–22
Market · NCES

85%
of 61 business schools require an IS course in the core
Market · Fustos et al., 2017

63
course documents from 31 institutions
Syllabi and institutions

52.9%
of 855 U.S. data analyst postings ask for SQL
Employment · 365 Data Science, 2026

Plus a **textbook study** (60 titles catalogued; four leading books compared chapter by chapter) and **classroom testing** with about 600 students in summer and fall 2026.

Degree and course figures measure audience scale and course presence, not this book's enrollments. Sources: NCES Digest of Education Statistics (2023), table 318.20; Fustos, Morris and Haga (2017), *Business and Management Studies* 3(2); 365 Data Science (2026), 855 relevant postings from 1,000 Indeed U.S. listings; author's syllabus and textbook audits. Full methods and references online.

The proposed edition: 16 chapters in five parts

A working proposal, open to better ideas. Plus a preface, a midterm Integration Studio, a closing chapter on careers and seven appendices.

PART I From Data to Decisions

1 **From Data to Business Performance**
Business, KPIs, DIKW, information behavior, IS and IT

2 **Understanding Data**
Data types, measurement scales, CSV and JSON, data quality

3 **Organizing Data and Information**
Files, metadata, taxonomies, data dictionaries

PART II Databases and SQL

4 **Databases and the Relational Model**
DBMS, keys, relationships, schemas, integrity

5 **SQL Foundations**
SELECT to aggregates, CREATE and INSERT, first AI verification

6 **SQL for Business Analysis**
Joins, GROUP BY, CASE, business measures

MID **Integration Studio A**
Review, practice exam, project checkpoint 1

PART III Designing and Running Databases

7 **Database Design: From Requirements to ER Diagrams**
Business rules, Crow's Foot notation, ERD tools

8 **Normalization**
Anomalies, dependencies, 1NF to 3NF

9 **Building and Testing a Database**
DDL, constraints, loading data, integrity tests

10 **Advanced SQL**
CTEs, views, window functions, triggers

11 **Database Administration, Security and Reliability**
Roles, row-level security, transactions, backup

PART IV From Data to Insight

12 **Business Analytics and Decision-Making**
Fair comparisons, causation, the decision memo

13 **Reporting and Data Visualization**
Dashboards, chart choice, honest visuals

14 **Power BI**
Power Query, data model, DAX measures, validated against SQL

15 **Big Data, Data Warehouses and AI**
Star schemas, ETL, NoSQL, verifying AI output

PART V Putting It to Work

16 **Capstone: From Data to Decisions**
Problem to database to dashboard to decision memo

END **Closing: Where Data Can Take You**
Data careers and building a portfolio

APPENDICES Tools and routes

A Supabase and the in-book SQL editor · B MySQL route (planned) · C Microsoft Access route · D SQL reference and dialect guide · E Diagramming · F Tableau route · G Terms Treasury

Every chapter: introduction, core concepts, Terms Treasury, a Reading Assessment Test, a Let's Build, an interactive lab, review questions and key takeaways, plus a three-checkpoint semester project.

Two routes: business core, Chapters 1–8 and 12–14; MIS, Chapters 1 and 4–16, with 2–3 as reading. A 14-week schedule is proposed; pacing to be agreed in review.

Available now

- Interactive textbook and reader
- A Let's Build, a lab and a quiz in every chapter
- Instructor dashboard and materials
- Access route; Grading Database and PetVax cases

Planned for the next edition

- The 16-chapter edition above
- In-browser PostgreSQL practice and a hosted Supabase project
- Power BI with a Tableau alternative; a MySQL route to test
- Automated SQL and lab feedback (prototype in testing)

Three decisions to make together

1
Publishing arrangement and portfolio positioning

2
Development scope: tools, cases and a sample unit

3
Responsibilities, review milestones and manuscript submission

I would like to formalize our publishing relationship and agree on the development scope and manuscript submission timeline.

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Full research, outline and references: dimapub.com/using-data-prospectus · Live reader: database-textbook.dimapublishing.com